



LECTURE 3

MACHINE LEARNING ON DATA REPRESENTATIONS

Learning on fixed representations

PRESS – OR SPACE. →

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Learning Outcome Roadmap

Learning outcome	Lecture	Block
☐ Explain how heterogeneous datasets align to a common coordinate support and form a geospatial data cube.	Lecture 3	Geospatial Data
☐ Represent locations as fixed feature vectors extracted from a geospatial data cube.	Lecture 3	Geospatial Data
☐ Explain how linear regression maps a fixed representation to a continuous target.	Lecture 3	Linear Models
☐ Explain how logistic regression maps a fixed representation to categories with a linear decision boundary.	Lecture 3	Linear Models
☐ Explain how decision trees partition a fixed representation space into nonlinear decision regions.	Lecture 3	Decision Trees
☐ Explain how random forests combine trees for robust predictions from fixed representations.	Lecture 3	Decision Trees

Many heterogeneous geospatial datasets become one consistent representation.

Geospatial Data Cubes

A fixed representation becomes input to a simple parametric mapping.

Linear Models

The same fixed representation supports nonlinear predictive rules.

Decision Trees & Random Forests

Fixed and Learned Representations

LECTURE 3

Construct representation → learn prediction

We align heterogeneous data, extract fixed features, and fit a predictive model.

LECTURES 4–5

Learn representation → learn prediction

Deep models learn useful internal features directly from data.

Lecture 3 learns on a representation. Lectures 4–5 learn the representation itself.